

FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), farmers, farm managers, and system administrators can use a centralized Salesforce-based platform to record, organize, and manage information related to farms, crops, farmers, and buyers. The system allows users to enter and update farm details, crop production records, and buyer transactions through a user-friendly interface supported by validation rules that help prevent incorrect or incomplete entries. It also provides real-time reports and interactive dashboards for analyzing farm productivity, monitoring sales, and supporting informed decisions. Role-based access and secure data storage help protect sensitive information while allowing effective collaboration among users. Overall, the system reduces manual paperwork, minimizes errors, improves operational efficiency, and supports data-driven farm management.

1. INTRODUCTION

1.1 Project Overview:

The Farm Management System (FMS) is a Salesforce-based application developed to digitize and simplify agricultural operations. It manages information associated with farms, crops, farmers, and buyers and provides real-time insights through reports and dashboards. The system enables farmers and farm managers to monitor agricultural activities efficiently and use the available information to make informed decisions.

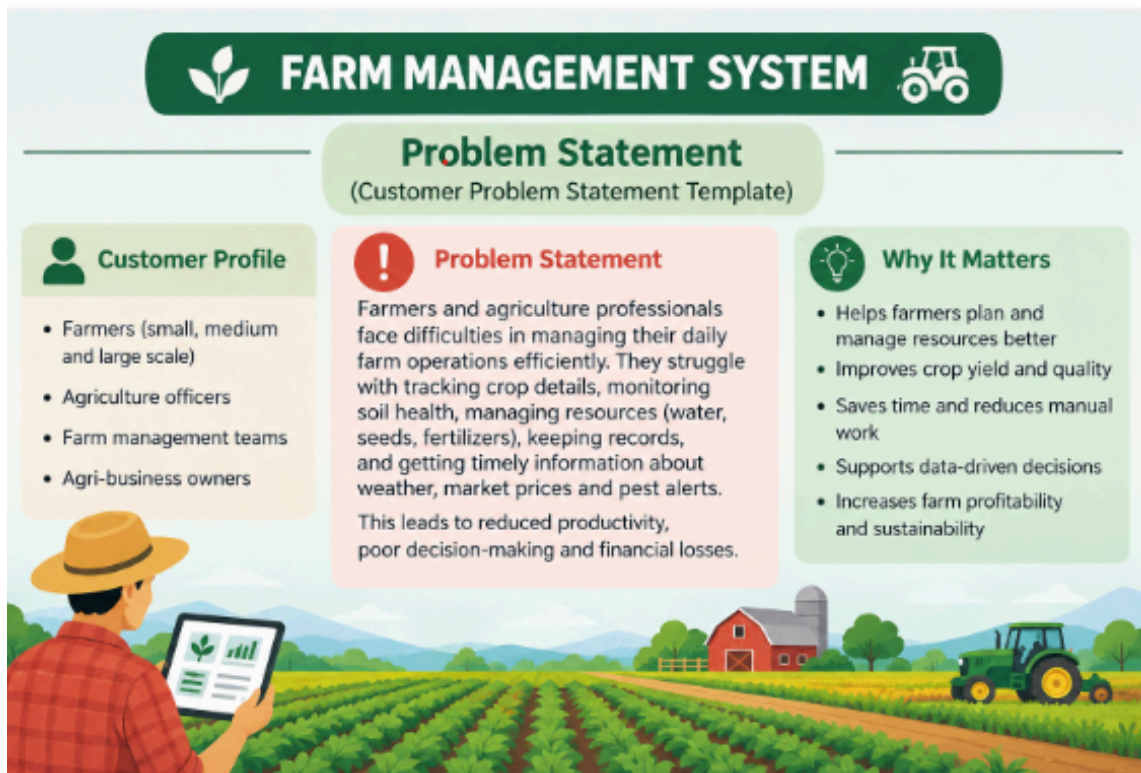
1.2 Purpose:

The primary objectives of this project are to:

- Digitize the management of agricultural data
- Improve the accuracy and accessibility of information
- Automate the handling of records
- Provide real-time monitoring capabilities
- Support farming decisions based on data
- Improve productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)



2.2 Empathy Map Canvas

Target Users:

- Farmers
- Farm Managers
- Buyers
- System Administrators
- Empathy Analysis
- Think & Feel: Require accurate and timely agricultural information
- Hear: Receive feedback from agricultural officers
- See: Physical registers, paper files, and spreadsheets
- Say & Do: Manually record and update information
- Pain: Risk of data loss, errors, and delays
- Gain: Automated processes and faster reporting

2.3 Brainstorming

- Ideas Generated During Brainstorming
- Manual maintenance of farm records
- Management using Excel spreadsheets
- Web-based farm management system
- Mobile-based farm application
- Salesforce-based FMS
- Selected Solution
- Salesforce-based Farm Management System.

3. REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Login
- Open the Farm Management Application
- Enter farm and crop information
- Register farmer and buyer details
- Generate required reports
- Analyze the generated results
- Make informed decisions

3.2 Solution Requirements

Functional Requirements



Non-Functional Requirements



NON-FUNCTIONAL REQUIREMENTS



(Farm Management System)



Performance

- System should respond within a reasonable time (fast loading and data processing).
- Should handle multiple users simultaneously without delay.



Security

- Protects user data and farm records.
- Uses secure login and authentication (e.g., password, role-based access).
- Prevents unauthorized access and data breaches.



Usability

- Simple and easy-to-use interface.
- Clear navigation and labels.
- Accessible for farmers with basic computer or mobile skills.



Reliability

- System should work correctly and consistently.
- Minimal downtime and errors.
- Data should be saved safely and not lost.



Scalability

- Should support more users, data and features in the future.
- Capable of handling increased farm operations and records.



Maintainability

- Easy to update and fix issues.
- Well-organized and documented code.
- Modular design for simple changes.



Portability

- Should work on different devices (desktop, laptop, mobile).
- Compatible with major operating systems and browsers.



Compatibility

- Integrates with existing systems and databases.
- Works with standard web technologies (HTML, CSS, JavaScript).



Non-Functional Requirements



3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Salesforce Validation Rules → Record Relationships → Reports → Dashboards

3.4 Technology Stack

Layer	Technology
Frontend	Salesforce Lightning UI
Backend	Salesforce Platform
Database	Salesforce Cloud
Automation	Salesforce Validation Rules
Analytics	Salesforce Reports and Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manually maintained farming records with an automated data management approach. Cloud storage helps reduce the risk of data loss, dashboards support better analysis, and real-time reports reduce reporting delays. Together, these capabilities provide accurate and timely agricultural information.

4.2 Proposed Solution

Parameter	Description
Problem	Manual maintenance of farm, crop, farmer, and buyer records can result in errors, data loss, and delays.
Solution	A Salesforce-based Farm Management System that provides centralized data storage and automated processes.
Innovation	Automated data validation, cloud-based storage, and real-time reporting through dashboards.
Impact	Improved data accuracy, faster reporting, and better decision-making for farmers and farm managers.
Business Model	A digital platform for farm management and agricultural data services.
Scalability	Supports multiple farms, users, and regions through cloud-based infrastructure.

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4.3 Solution Architecture

Architecture Layers

Presentation Layer – Salesforce Lightning UI

Application Layer – Salesforce Validation Rules, Relationships

Data Layer – Custom Objects

Analytics Layer – Reports and Dashboards

PROJECT DEVELOPMENT PHASE

Project Description:

The Farm Management System (FMS) developed in Salesforce is a comprehensive solution for digitizing and streamlining agricultural operations. The application supports farmers, farm managers, and other stakeholders by managing information related to crops, farms, farmers, and buyers. It also provides useful insights through reports and dashboards.

Milestone 1 – Salesforce:

Milestone 2 – Objects:

Milestone 3 -Tab:

Milestone 4 – Lightning App:

Milestone 5 – Fields and Relationships:

Milestone 6 – Profile:

Milestone 7 – Role:

Milestone 8 – User:

Milestone 9 – Sharing Rules:

Milestone 11 – Reports:

Milestone 12 – Dashboard:

Milestone 13 – View Report and Dashboard:

Milestone 1 – Salesforce:

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to <https://developer.salesforce.com/signup>
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email



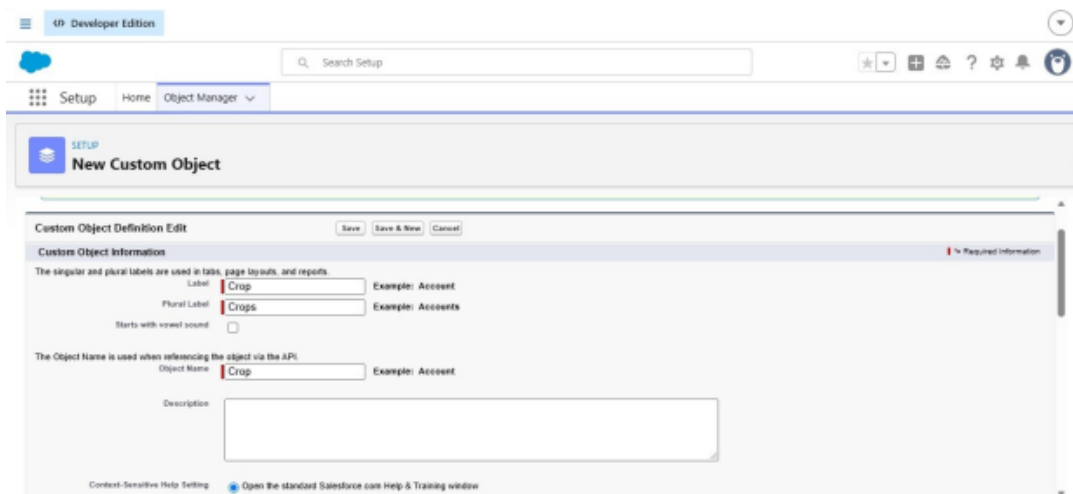
Milestone 2 – Objects:

Activity 1:

Create a custom object for Farm:

Follow these steps to create a custom object:

1. From setup click on object manager.
2. Click create, select custom object.
3. Enter the label as "Farm".
4. Enter the plural label as "Farms".
5. Record Name: "Farm Name"



Activity 2:

Creating the Crop object:

1. Click the gear icon and select Setup.

2. Select the Object Manager tab next to the Home tab.
3. On the right side, locate the Create dropdown and select Custom Object.

Activity 3:

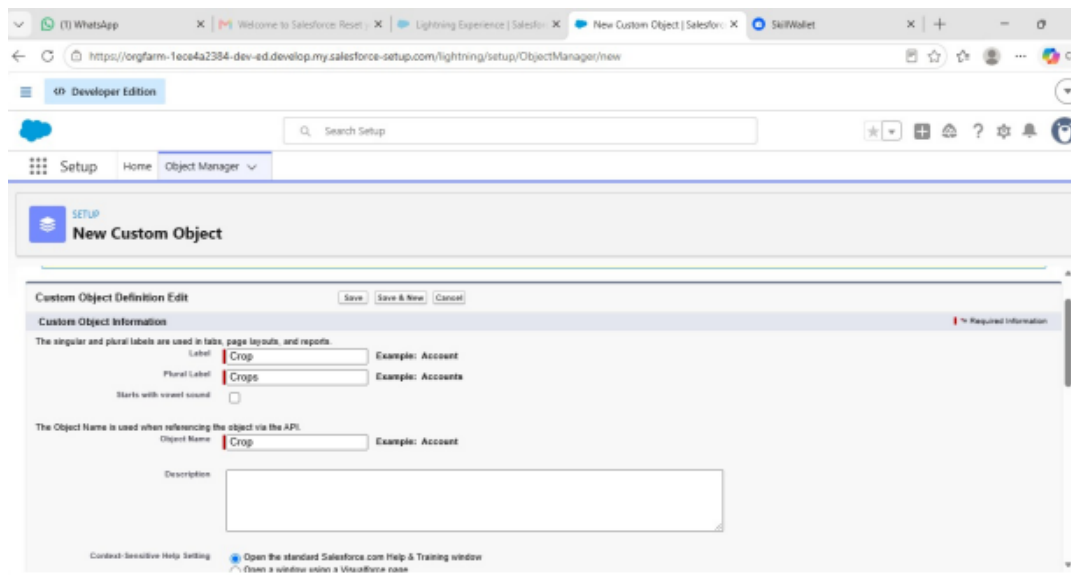
Creating the Farmer object:

1. Click the gear icon and select Setup.
2. Select the Object Manager tab next to the Home tab.
3. On the right side, locate the Create dropdown and select Custom Object.

Activity 4:

Creating the Buyer object:

1. Click the gear icon and select Setup.
2. Select the Object Manager tab next to the Home tab.
3. On the right side, locate the Create dropdown and select Custom Object.



Milestone 3 -Tab:

Activity 1:Creation of Farm tab:

Create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farm.
4. For Tab Style, choose an appropriate icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:Creation of Crop tab:

Create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Crop.
4. For Tab Style, choose an appropriate icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:Creation of Farmer tab:

Create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Farmer.
4. For Tab Style, choose an appropriate icon.
5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:Creation of Buyer tab:

Create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, choose an appropriate icon.
5. Leave all defaults as is. Click Next, Next, and Save.

WhatsApp

Welcome to Salesforce: Reset

Lightning Experience | Salesforce

Tabs | Salesforce

SafariWallet

https://orgfarm-1ccc4a2384-dev-ed.develop.my.salesforce-setup.com/lightning/setup/CustomTabs/page?address=%2Fsetup%2Fu%2Fcustomtabs.jsp%3Fsetu...

Developer Edition

Search Setup

SetupHomeObject Manager

tabs

User Interface

Rename Tabs and Labels

Tabs

Didn't find what you're looking for? Try using Global Search.

SETUP

Tabs

Custom Object tabs look and behave like the standard tabs provided with Salesforce. Web tabs allow you to embed external web applications and content within the Salesforce window. Visualforce tabs allow you to embed Visualforce pages. Lightning Component tabs allow you to add Lightning components to the navigation menu in Lightning Experience and the mobile app. Lightning Page tabs allow you to add Lightning Pages to Lightning Experience and the mobile app.

Custom Object Tabs

NewWhat Is This?

Action	Label	Tab Style	Description
Edit Del	Budget	Full	
Edit Del	Goods	Globe	
Edit Del	Expenses	Globe	
Edit Del	Assets	Globe	

Web Tabs

NewWhat Is This?

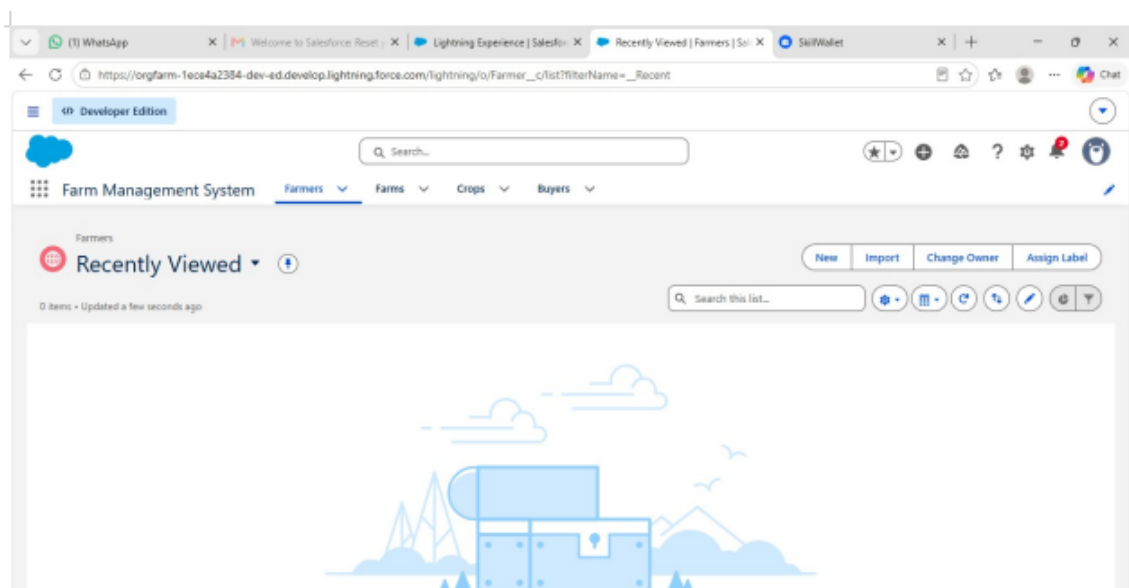
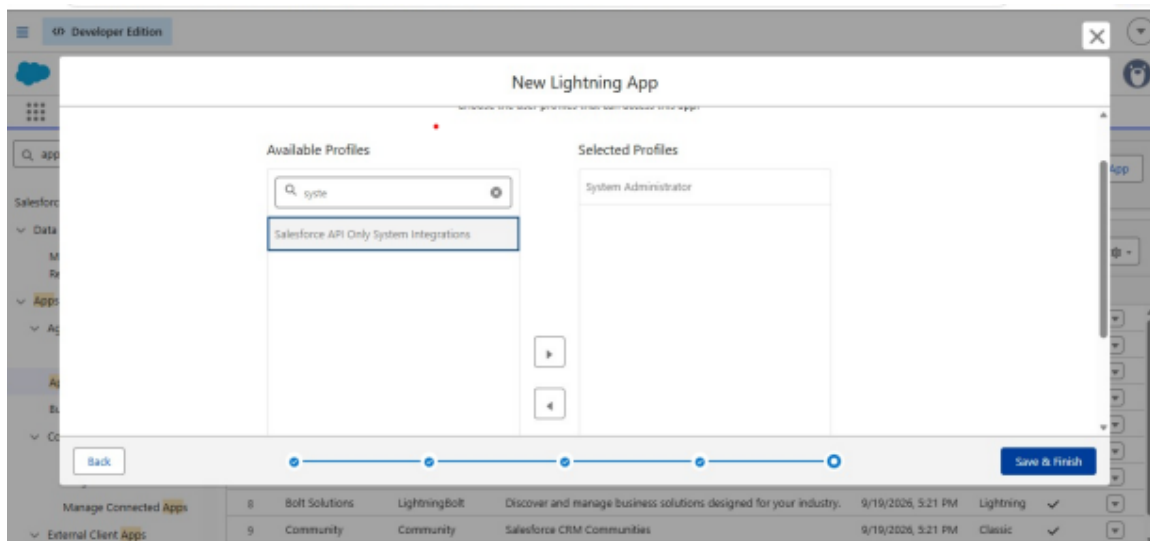
No Web Tabs have been defined

Visualforce Tabs

NewWhat Is This?

Milestone 4 – Lightning App:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.
3. Navigation menu-Displays the tabs available inside the app.



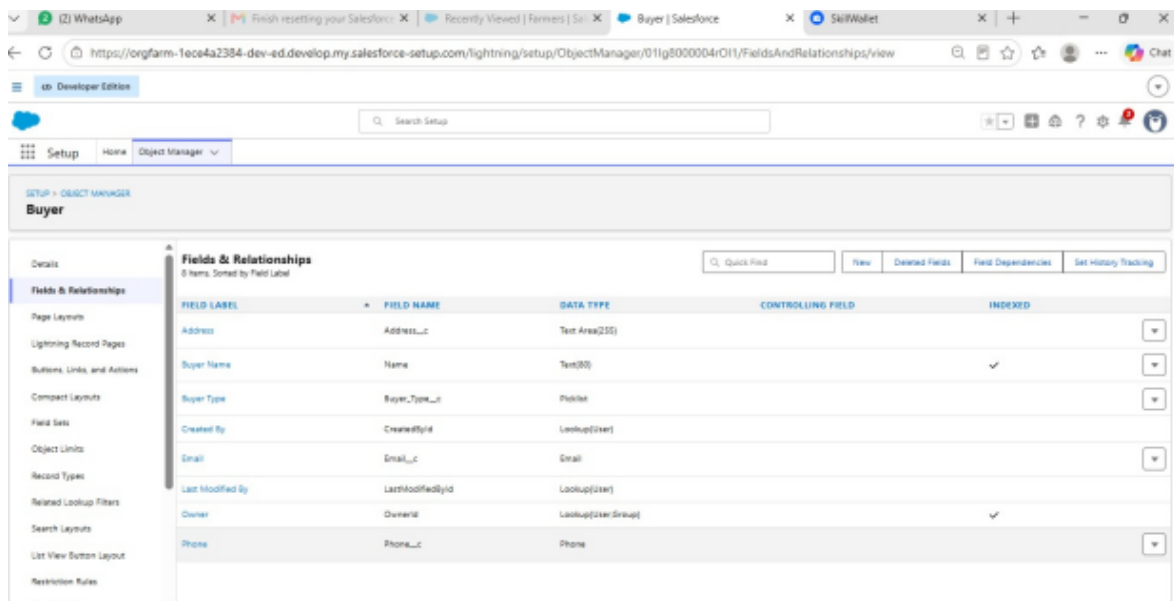
Milestone 5 – Fields and Relationships:

Activity 1: Creating fields for the Farm object:

Activity 2: Creating fields for the Crop object:

Activity 3: Creating fields for the Farmer object

Activity 4: Creating fields for the Buyer object:



Milestone 6 – Profile:

Activity 1: Create a custom profile:

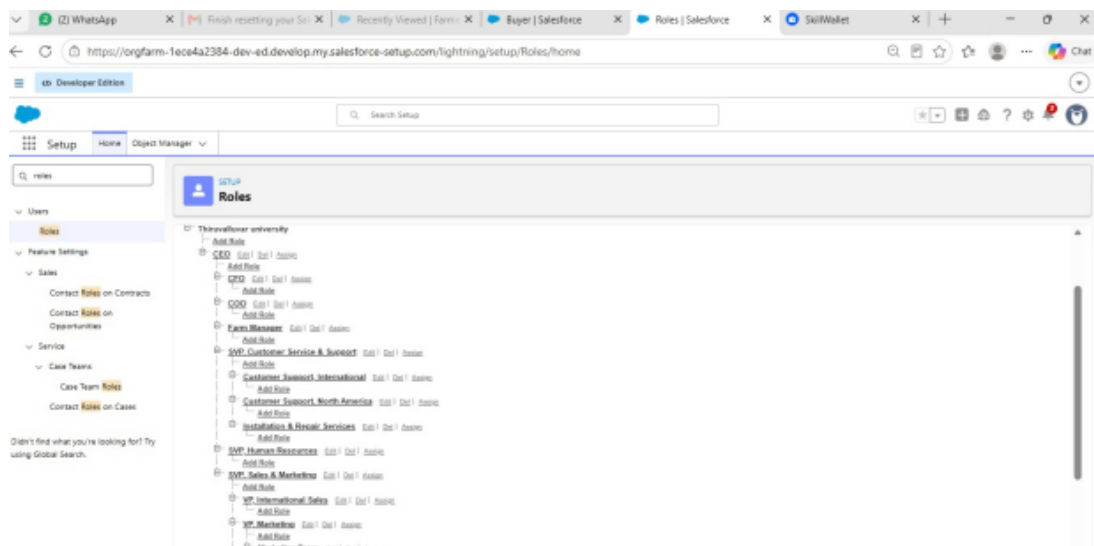
Activity 2: Create a second custom profile:

Milestone 7 – Role:

Activity-1

Creating a role

1. Use Quick Find to search for Roles and select the Roles option.
2. Select the Set Up Roles option.
3. Under CEO, click Add Role and enter "Farm Manager" as the label. The role name will be populated automatically. Click Save.



FUNCTIONAL & PERFORMANCE TESTING

- During the testing phase, screenshots were captured from the Salesforce interface to confirm system functionality. The following areas were tested:
- Creation of Farm, Crop, Farmer, and Buyer records
- Validation of input fields
- Verification of relationships and field dependencies
- Testing of profile and role-based access
- Verification of sharing rules
- Generation of reports and dashboards
- Performance testing was carried out to verify smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data management
- Reduced dependence on paperwork
- Improved data accuracy
- Improved monitoring capabilities
- Real-time analytical insights

Disadvantages

- ☐ Requires an internet connection
- ☐ Depends on Salesforce licensing
- ☐ Provides limited offline access

9. Conclusion:

The Farm Management System successfully digitizes agricultural data management through Salesforce. It improves operational efficiency, reduces errors, and supports farmers and managers in making informed decisions through automation, reporting, and analytics.

FUTURE SCOPE

- Integration with a mobile application
- AI-based crop prediction
- Integration of weather data
- Connectivity with IoT sensors
- Integration with government schemes

- Market price forecasting
- APPENDIX
- Source Code: Salesforce configuration and metadata
- Dataset: Farm, Crop, Farmer, and Buyer records
- Tools Used: Salesforce Developer Org